

LongDSL Night 1 Bootstrap Report

GIANT v3 synthetic long-context bootstrap experiment note.

Purpose

This document records the first autonomous LongDSL stress-test session for GIANT v3. The immediate goal was not to reach 64K or 1M context in one night, but to answer a narrower question first:

- Can a small GIANT v3 baseline learn the LongDSL retrieval task at all?
- Is direct-from-scratch long-context training viable, or is a short-context bootstrap required?
- Does the transfer path survive the first context jumps?

Experiment Summary

Model ~28.5M params 384 dim, 6 heads, 12 layers, FF 1536	GPU 1x RTX A6000 single GPU RunPod secure cloud session	Best Result 45.31% exact match level 1, 128 tokens, transfer from 64-token checkpoint
DSL Focus Level 1 retrieval entity -> value with aliasing and distractors	Best Bootstrap 64 -> 128 lower-LR transfer-only answer adaptation	Failure Point 256 tokens transfer quality collapsed sharply

LongDSL Setup

DSL structure

The new synthetic pipeline lives under GIANT/v3/Long. Three levels were implemented:

- Level 1: DEF, SET, ALIAS, ASK
- Level 2: INC, DEC, SWAP
- Level 3: DEFARR, SETAT, SWAPAT, INCAT, GET

The overnight runs stayed on level 1 only. That was intentional: the baseline first had to prove it could learn the simplest retrieval version before pushing to more complex operators or longer contexts.

Tokenization and data layout

- Small custom word-level tokenizer stored at /proj/giant-data/GIANT/Long/tokenizers/longdsl_wordlevel
- Raw JSONL samples stored under /proj/giant-data/GIANT/Long/raw/...
- Arrow shards stored under /proj/giant-data/GIANT/dataset_artifacts/...
- Curated saved night results stored under /proj/giant-data/GIANT/single-gpu/checkpoints/longdsl/night1

Baseline model

```
embedding_size: 384
num_heads: 6
num_kv_heads: 6
num_layers: 12
feed_forward_size: 1536
rope_dim: 64
dropout_rate: 0.0
```

param_dtype: float32
compute_dtype: bfloat16

This is the smallest GIANT v3 shape that looked reasonable while still being cheap enough to iterate on quickly.

Training Variants Tried

1. Direct long-context baseline

The first family of runs tried to learn the task directly at larger context lengths, including 512-token setups. Several dataset and masking variants were tried during the night:

- chat-style masked assistant target
- plain LM ending with ANS Vxxx
- two-stage LM -> answer-only supervision at the same context length

These direct 512-token baselines consistently stayed near floor on held-out exact match. This was the strongest negative result of the session.

2. Bootstrap curriculum

The first useful signal appeared only after switching to a short-context curriculum:

- bootstrap at 64 tokens
- transfer from the 64-token checkpoint into 128 tokens
- lower learning rate during transfer
- answer-focused adaptation instead of re-running the whole curriculum from scratch

This transfer path was clearly better than direct 512-token training.

Key Results

Held-out exact match

Direct baseline @ 512		4 / 128	baseline_v5, LM + answer-only at 512 from scratch
Bootstrap @ 64		56 / 128	bootstrap_v1, first positive result
64 -> 128 transfer		58 / 128	bootstrap_v2_transfer, best run of the night
128 -> 256 transfer		6 / 128	quality collapsed after the next context jump

Interpretation

- The task is learnable by the model family, so the idea is not dead.
- Direct-from-scratch long-context training is a poor entry point.
- A short-context bootstrap is necessary.
- The 64 -> 128 transition works.
- The 128 -> 256 transition does not yet hold, which means the curriculum is still too aggressive or too sparse.

What Actually Changed the Outcome

The main engineering lesson from the night is that the data formulation mattered more than any architectural tweak.

The final useful setup was:

- short-context bootstrap first
- lower-LR transfer after bootstrap
- answer-only supervision at transfer time
- reuse of the smaller-context checkpoint instead of restarting from scratch

The following approaches did **not** unlock learning:

- jumping straight to 512 tokens from scratch
- sparse answer-only training without a bootstrap stage
- retraining a larger context with a fresh optimizer and no careful transfer recipe

Log Shape Notes

The LM bootstrap stages usually converged to a narrow band around loss 2.3 - 2.6, which is enough to make the DSL syntax/model state somewhat legible to the model, but not enough by itself to solve retrieval.

The answer-only stages showed a very different signature:

- individual batches sometimes hit very low loss
- neighboring batches could still spike badly
- this means the model sometimes retrieved the correct value, but the behavior was not stable enough across the whole evaluation set

That pattern is exactly what the exact-match results show:

- retrieval skill appears locally
- robustness across samples collapses too early as context grows

Recommended Next Setup

The next experiment should **not** continue from 256 immediately. The evidence says a better curriculum is needed first.

Recommended next changes:

- Make level 1 easier during bootstrap:
 - shorter alias chains
 - lower distractor density
 - fewer active entities/values early on
- Keep the transfer recipe that worked:
 - bootstrap at a short length
 - lower-LR transfer-only answer adaptation
- Add finer ladder steps instead of big jumps:
 - 64 -> 96 -> 128 -> 192 -> 256
- Only retry 512 after 256 is clearly stable.

Main Conclusion

The overnight result is a partial win, not a headline win.

- We did **not** prove that the current 30M GIANT v3 setup can simply scale LongDSL retrieval to 256+ tokens by brute force.
- We **did** prove that the task becomes learnable once the model is bootstrapped at short context and transferred carefully.

- The correct next research direction is curriculum design and transfer stability, not immediate long-context scaling.

That is enough signal to justify a second experiment configuration, but not enough to claim that the current setup is ready for the long-context ladder.

Saved Artifacts

Important local artifacts:

- /proj/giant-data/GIANT/single-gpu/checkpoints/longdsl/night1/bootstrap_v1_l1_ctx64/step_0010240.npz
- /proj/giant-data/GIANT/single-gpu/checkpoints/longdsl/night1/bootstrap_v2_transfer_l1_ctx128/step_0016384.npz
- /proj/giant-data/GIANT/single-gpu/checkpoints/longdsl/night1/bootstrap_v2_transfer_l1_ctx256/step_0019456.npz
- /proj/giant-data/GIANT/single-gpu/checkpoints/longdsl/night1/baseline_v5_l1_ctx512/step_0005120.npz

Mirrored S3 location:

- s3://giant-data/GIANT/single-gpu/checkpoints/longdsl/night1/

Session Status

The GPU pod used for the overnight run was terminated at the end of the session after the result checkpoints and logs were copied back and synced.